

Keshav Dutt Kaushik – Generative AI Engineer



Profile Summary:

Keshav has spent close to 4 years at Cognizant, contributing to multiple AI-driven products and accelerators within Cognizant's QE&A practice. He began on the QEA Marketplace team, building backend tooling that supports automation and efficiency in software engineering processes, and has been part of the Machine Learning and Generative AI team at the QEA TechCoE for over 2 years.

His focus is on backend engineering for Generative AI products — designing microservice APIs, building document ingestion and parsing pipelines, and integrating Retrieval-Augmented Generation (RAG) workflows with vector databases. His recent work spans an LLM-powered Knowledge Management Chatbot, a modernized React-based AI application (AILA), and an in-progress Generative AI Assurance platform.

Primary Responsibilities:

- Designing and implementing FastAPI-based microservice backends for LLM-powered applications and chatbots
- Building document ingestion and parsing pipelines across PDF, DOCX, PPTX, XLSX, and Markdown formats
- Integrating vector databases (Qdrant, Milvus) into RAG architectures for semantic search and knowledge retrieval
- Developing embedding pipelines using sentence-transformers with chunking strategies via LangChain
- Securing API endpoints with OAuth2 authentication and applying caching mechanisms for latency optimization
- Migrating legacy backend services from Flask to FastAPI and remediating production data-quality issues
- Collaborating with frontend teams on API contracts and supporting cross-functional product delivery

Skills and Qualifications:

- Bachelor of Technology in Computer Science, Manipal University Jaipur
- Strong foundation in Python and Java, with hands-on backend development in FastAPI, Flask, and Spring Boot
- Expertise in Retrieval-Augmented Generation (RAG) architectures, vector databases (Qdrant, Milvus), embedding models (sentence-transformers), and orchestration frameworks (LangChain)
- Experience with multi-format document parsing (Apache PDFBox, Python parsing libraries) to build AI knowledge bases
- Familiar with API security patterns (OAuth2), performance optimization through caching, and microservice architectures
- Proficient in legacy-code migration (Flask → FastAPI, legacy React → modern React) and refactoring within large enterprise codebases
- Strong problem-solving skills with experience shipping production AI features end-to-end